Significant Digits

The certainty of a measurement is communicated by the number of significant digits in the measurement.

Rules:

1) If a decimal point is present, zeros to the left of the first non-zero digit (leading zeros) are not significant 0.000$\boxed{81}$

2) If a decimal point is not present, zeros to the right of the last non-zero digit (trailing zeros) are not significant

3) All other digits are significant

4) When a measurement is in scientific notation, all digits in the coefficients are significant

5) Counted and defined values have infinite significant digits

Measurement	Sig Figs
32.07 m	4
0.0041 g	2
6400 s	2
10.0 KJ	3
100 people (counted)	∞

When multiplying or dividing, the answer is limited to the lower number of significant digits of the terms being multiplied or divided.

$\overset{3}{77.8}$ km/$\cancel{h}$ × $\overset{4}{0.8967}$ $\cancel{h}$ = 69.76326

69.8

6.98×10 km

$\frac{\overset{2}{35000} \text{ mol}}{\underset{3}{1.20} \text{ L}}$ = 29166.6667

2.9×10^4 mol/L

2.95 → 3.0

2.85 → 2.8

Rounding

If you're rounding from a number below 5 ROUND DOWN

If you're rounding from a number above 5 ROUND UP

If you're rounding from 5 ROUND TO THE EVEN

85 → 80

0.004395 → 0.00440

Ex) Round to the nearest integer

a) 1.1 1

b) 1.6 2

c) 1.5 2

d) 2.5 2

e) 2.5137 3